



UNDERSTANDING



VISUALIZING



COMMUNICATING

OPEN SPACE & 311 REQUESTS

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ARTG2242



UNDERSTANDING

Data Profile

Open Space Dataset:

publicly designated open spaces, including parks, playgrounds, protected land, and other green or recreational areas.

311 Request Dataset:

a continuously updated log of every service request, covering everything from potholes to missed trash pickups, tracking what was requested, where, when, and whether it was resolved.



Persona

Lindsey Matthews
Urban Planner

Agenda and Goal

- Access to quality open space within a walkable distance
- Build community trust through transparent, inclusive planning processes



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Persona

Lindsey Matthews
Urban Planner

Needs

- Understand whether the distribution is improving or worsening over time
- Understand where to direct attention and funding



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My Dashboard



BOSTON OPEN SPACE/311 SERVICE REQUESTS

Open Space Metadata: Originally released in October 2014, last modified in May 2024, Available in multiple formats (GeoJSON, CSV, Shapefile, KML).

311 Requests Metadata: The dataset is mid-transition to a new backend system, with full migration expected by mid-2026.

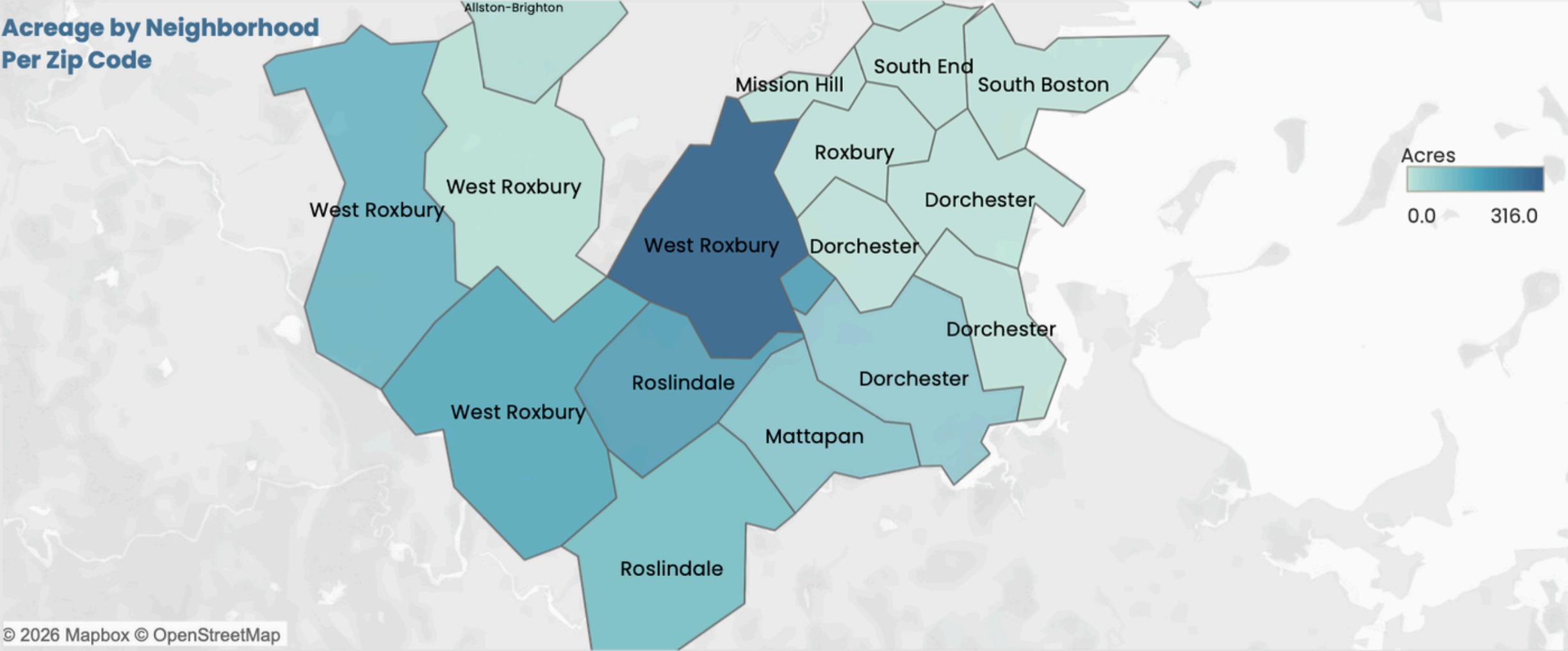
2025 ▼

Total 311 Requests:
50,265

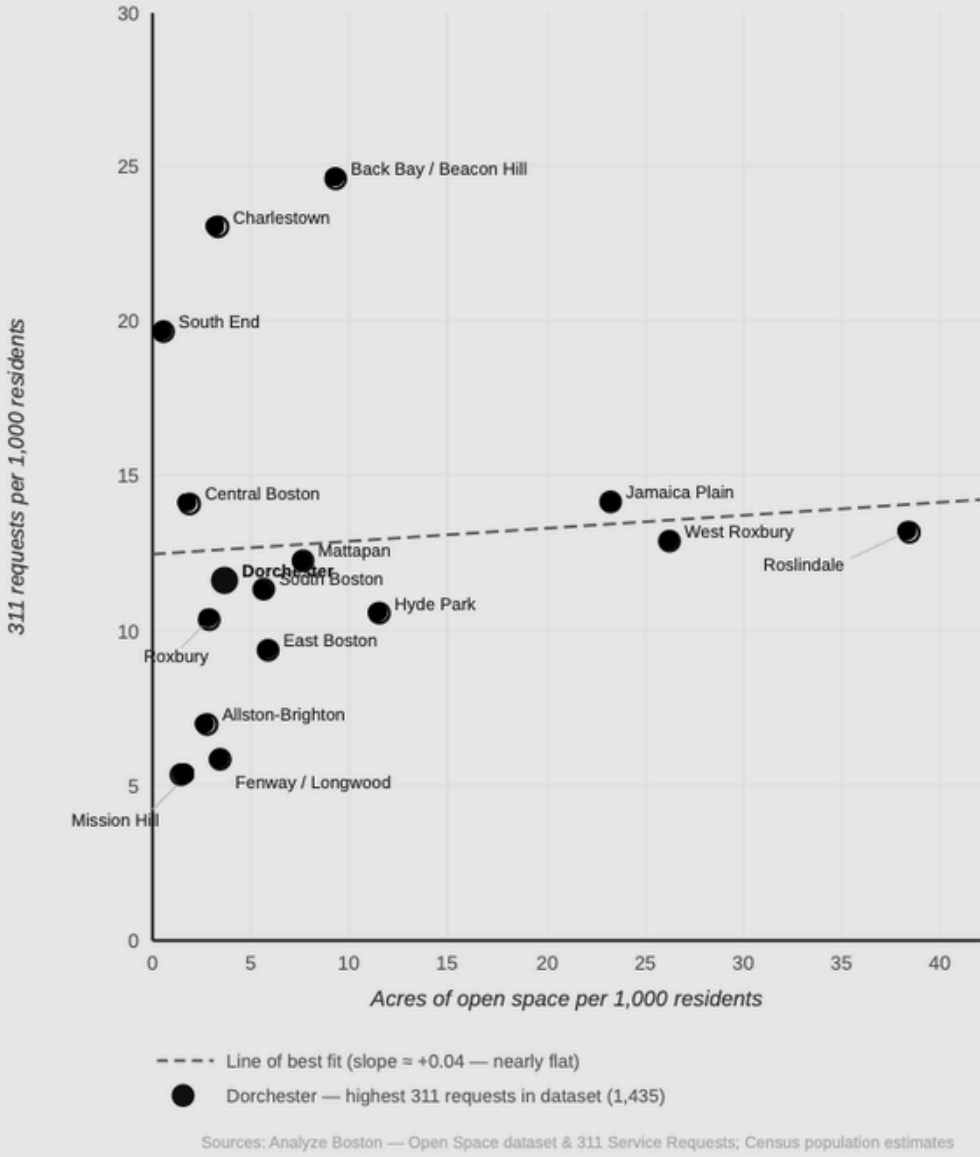
Most Common Department
Public Works

% of 311 Requests
On Time
70.5%

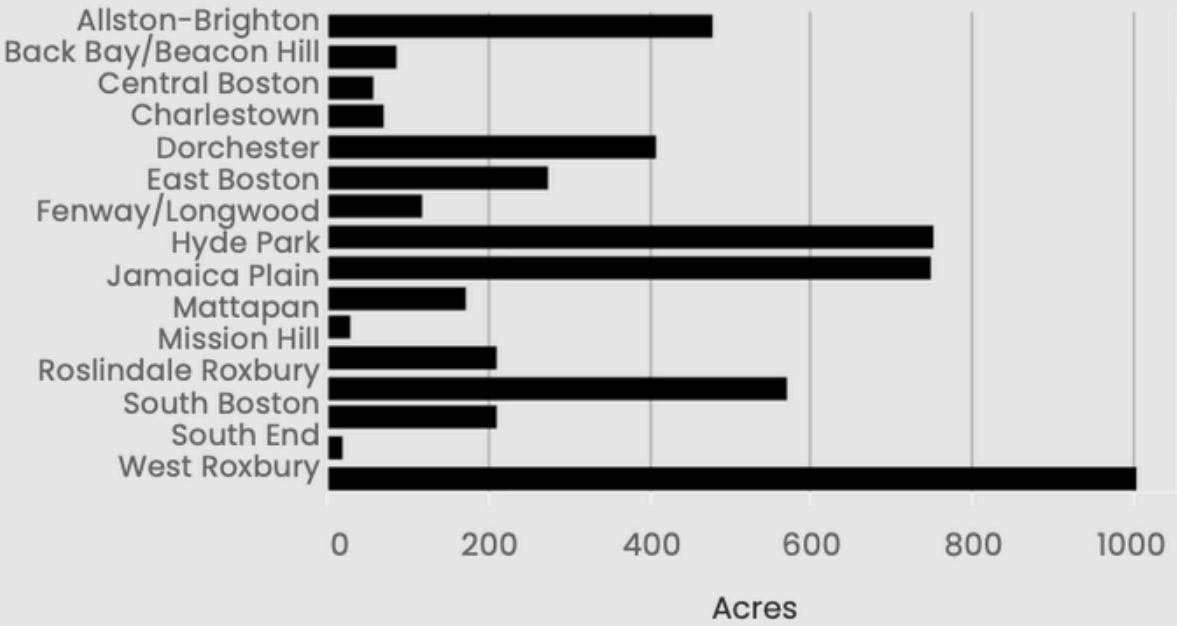
Acreage by Neighborhood
Per Zip Code



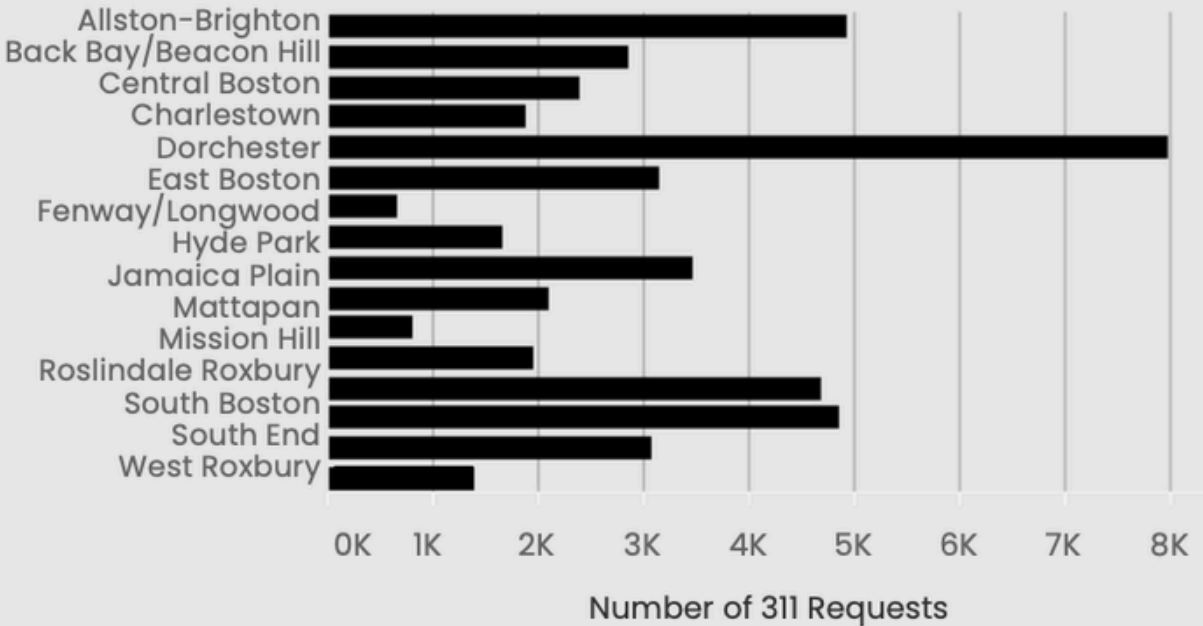
Open Space vs 311 Demand by
Neighborhood per Capita



Total Open Space per Neighborhood



311 Service Requests per Neighborhood



BOSTON OPEN SPACE/311 SERVICE REQUESTS

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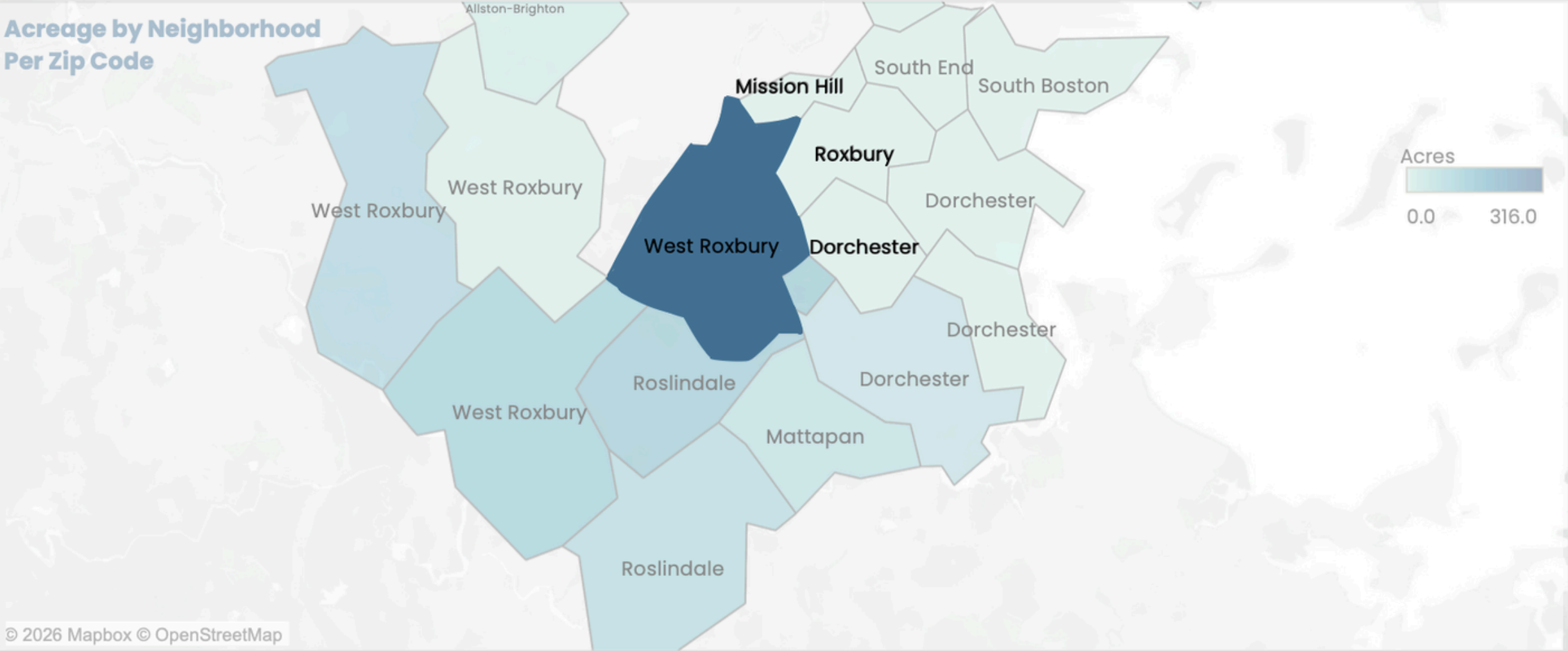
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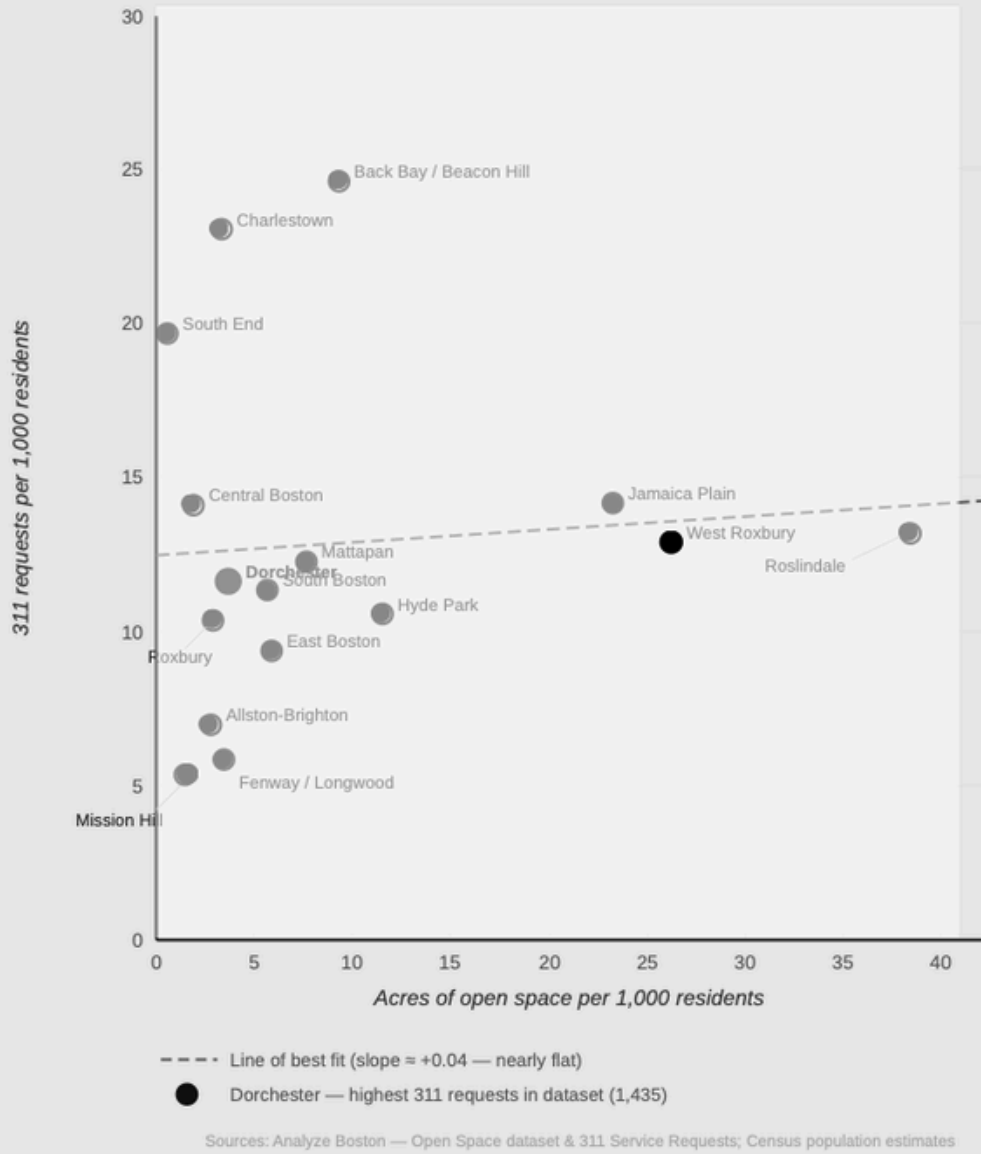
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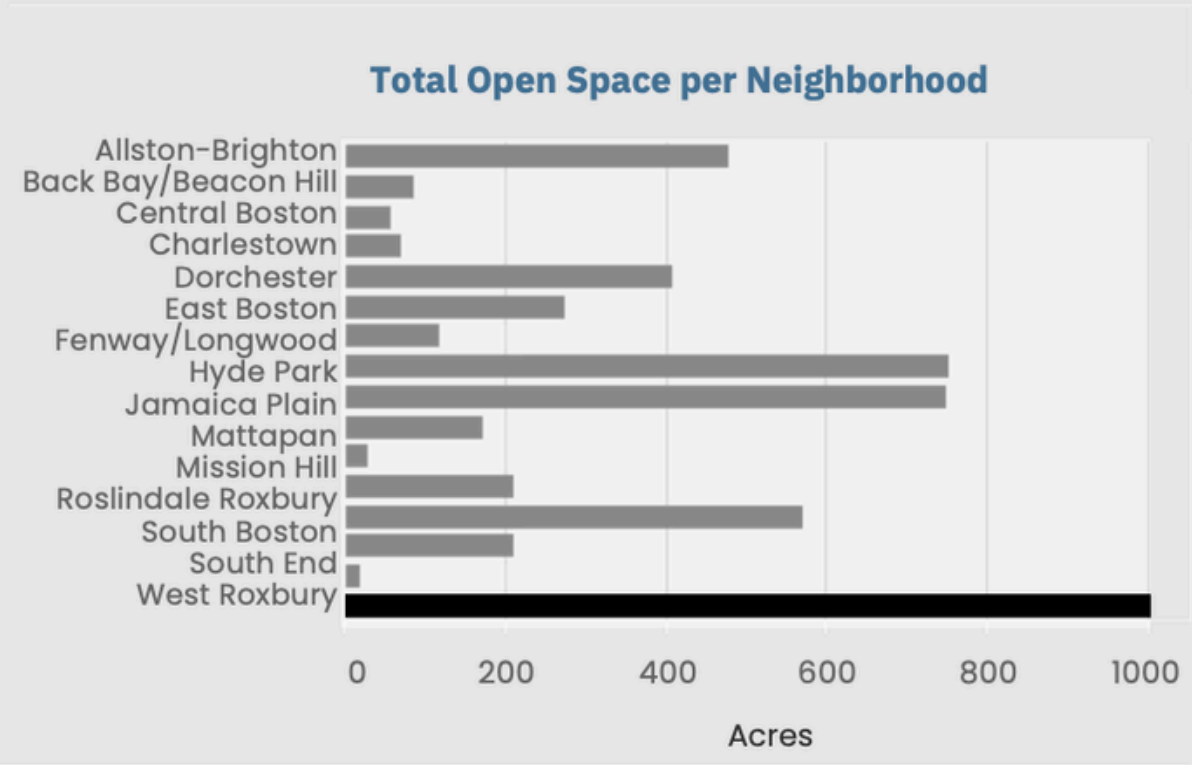
Acreage by Neighborhood
Per Zip Code



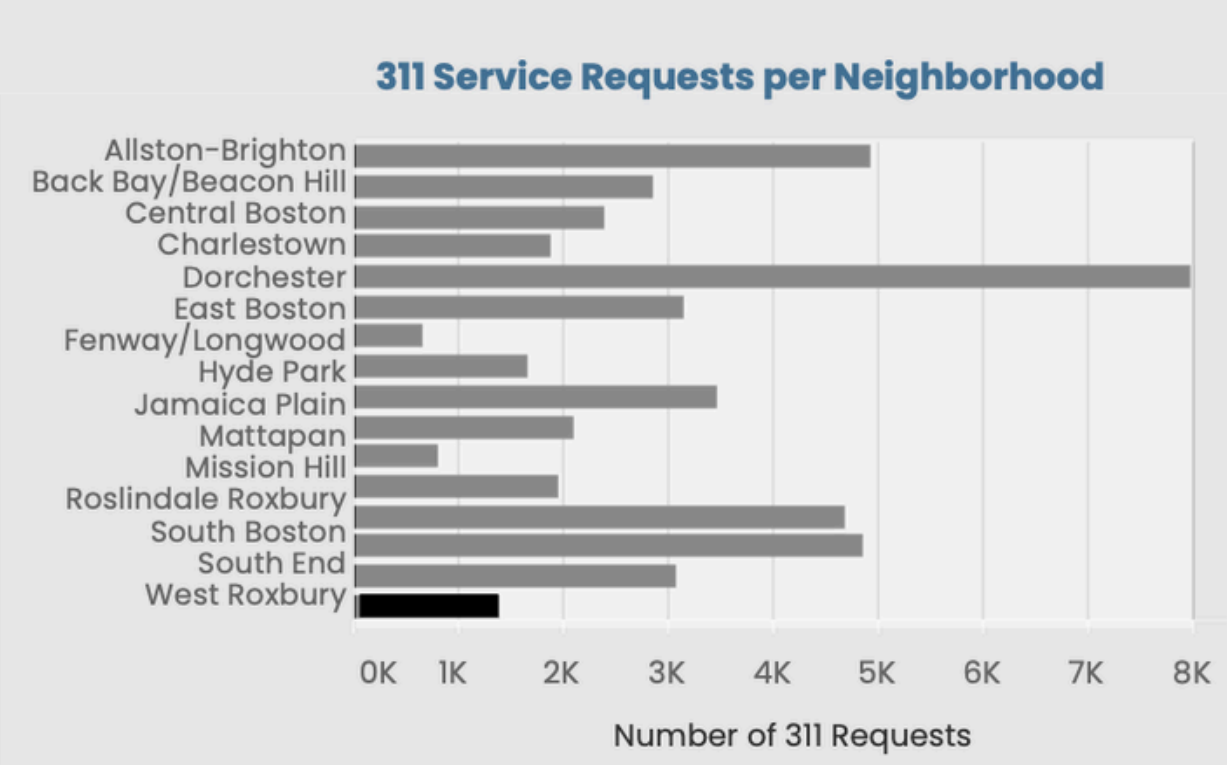
Open Space vs 311 Demand by
Neighborhood per Capita



Total Open Space per Neighborhood



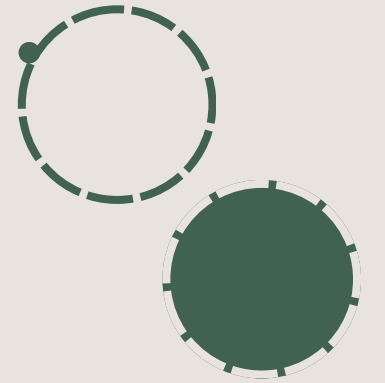
311 Service Requests per Neighborhood



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KPI Cards:

Total Requests, Most Common Dept, % On Time



Total 311 Requests:

50,265

% of 311 Requests On Time

70.5%

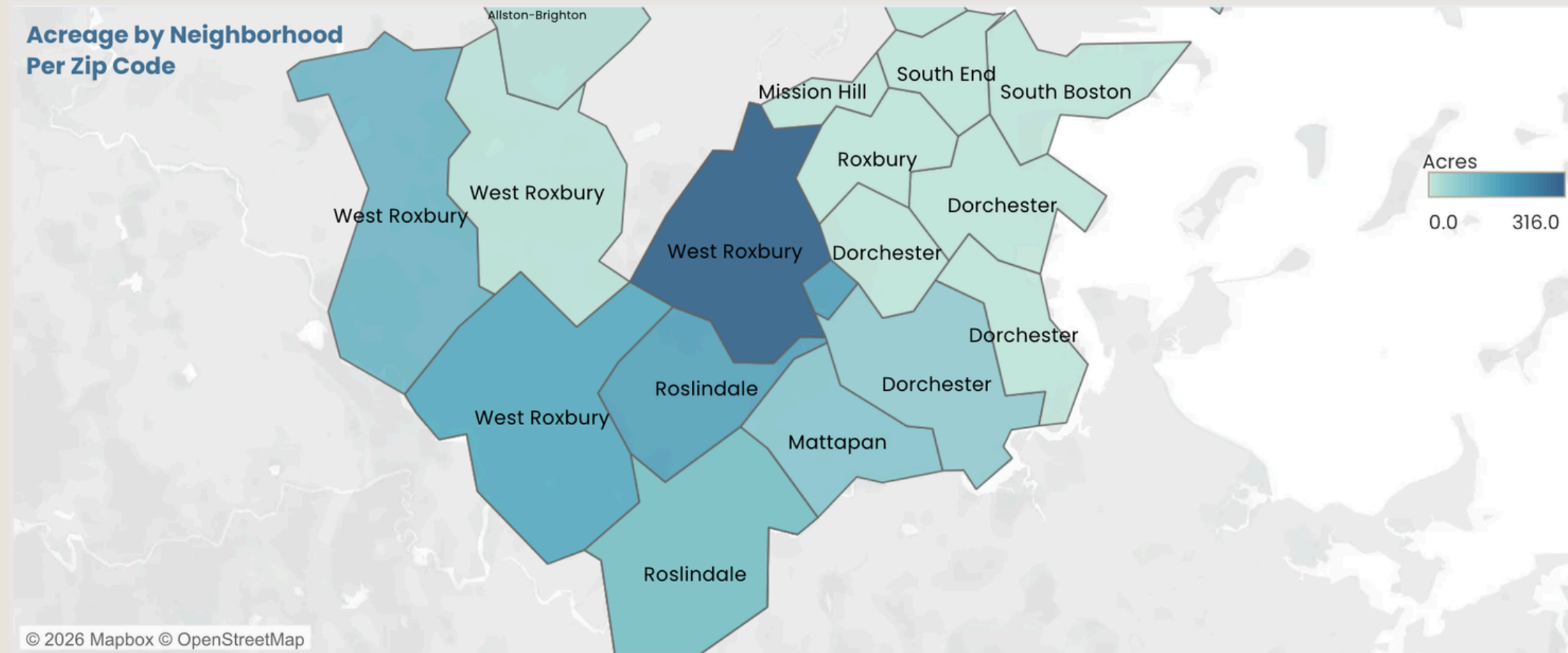
Most Common Department

Public Works



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Chloropleth Map: Acreage by Neighborhood Per Zip Code

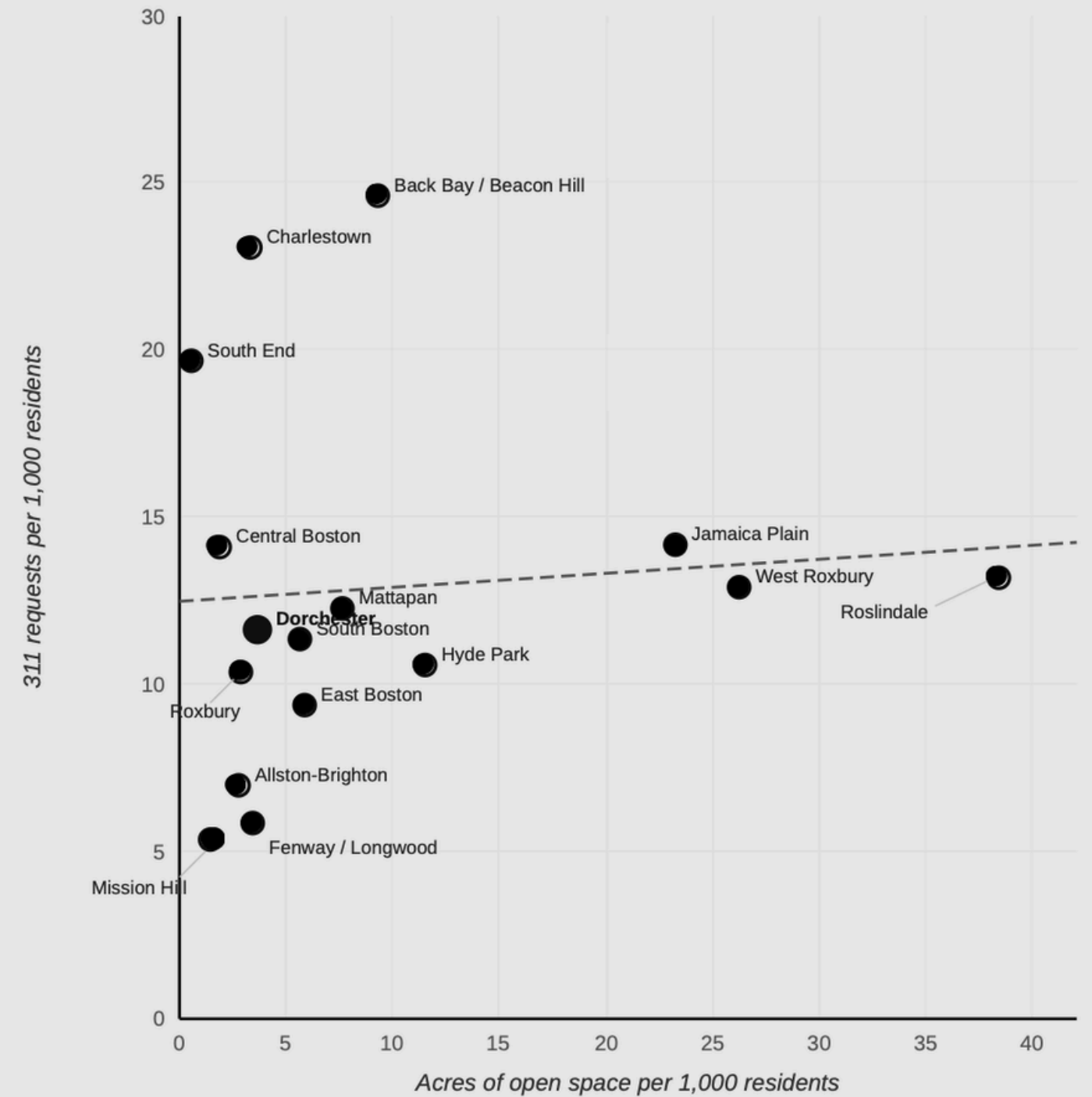


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Scatter Plot: Open Space vs. 311 Demand by Neighborhood per Capita



Open Space vs 311 Demand by
Neighborhood per Capita



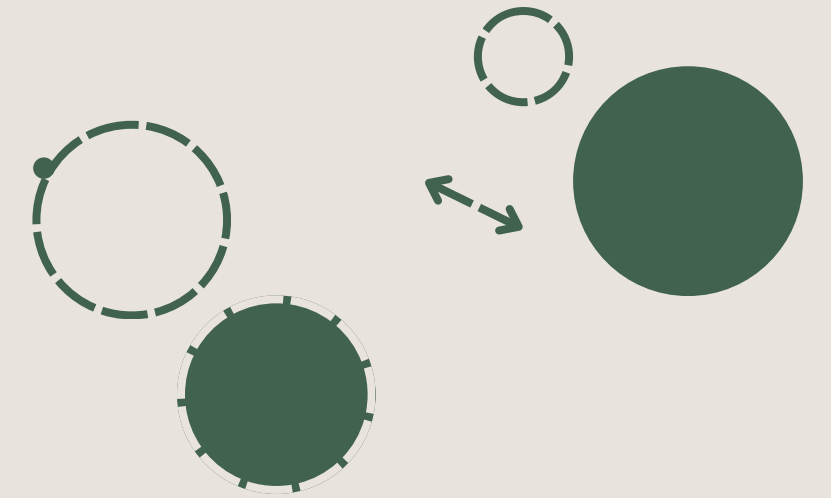
--- Line of best fit (slope $\approx +0.04$ — nearly flat)
● Dorchester — highest 311 requests in dataset (1,435)

Sources: Analyze Boston — Open Space dataset & 311 Service Requests; Census population estimates

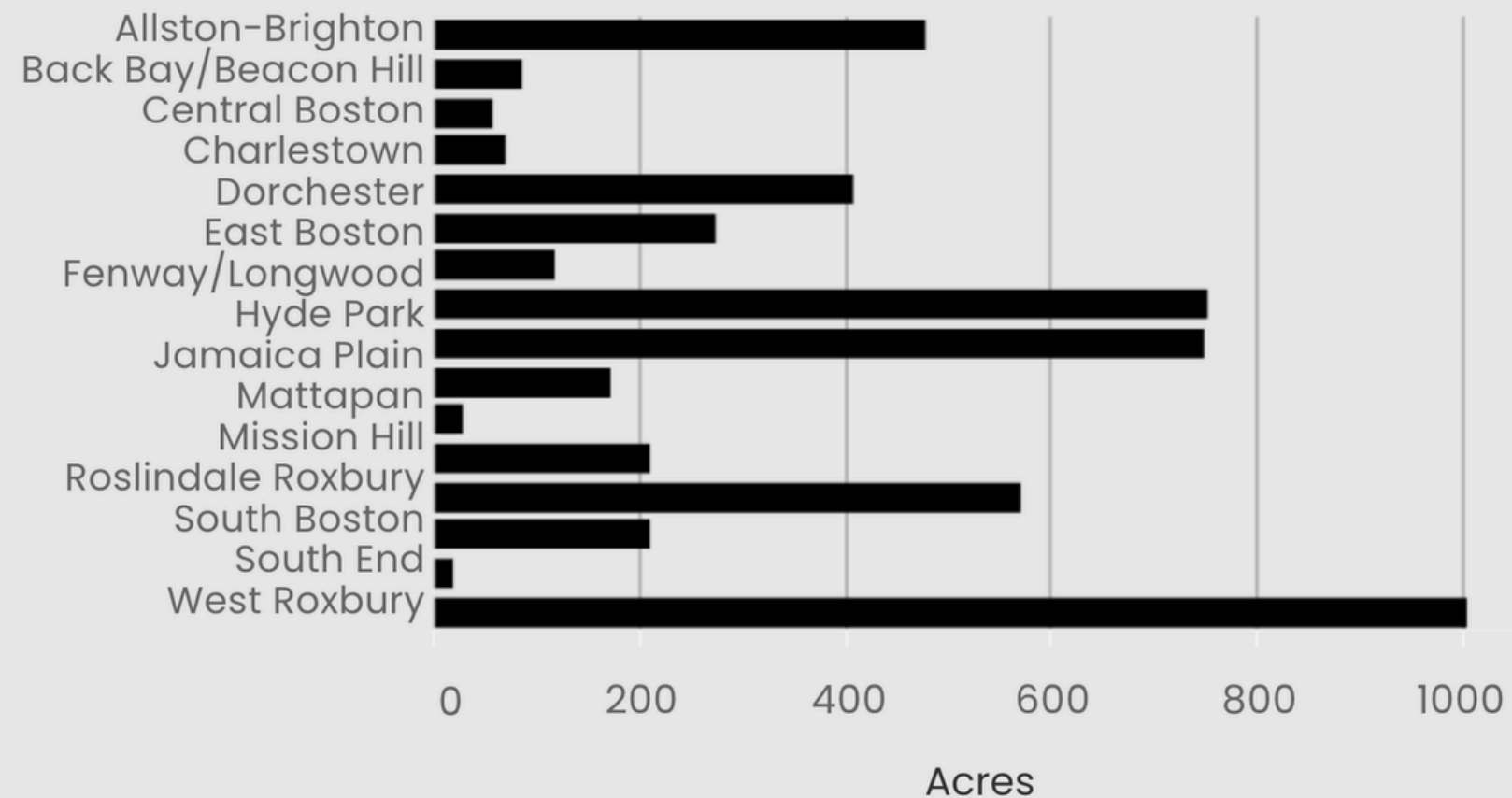
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Bar Chart:

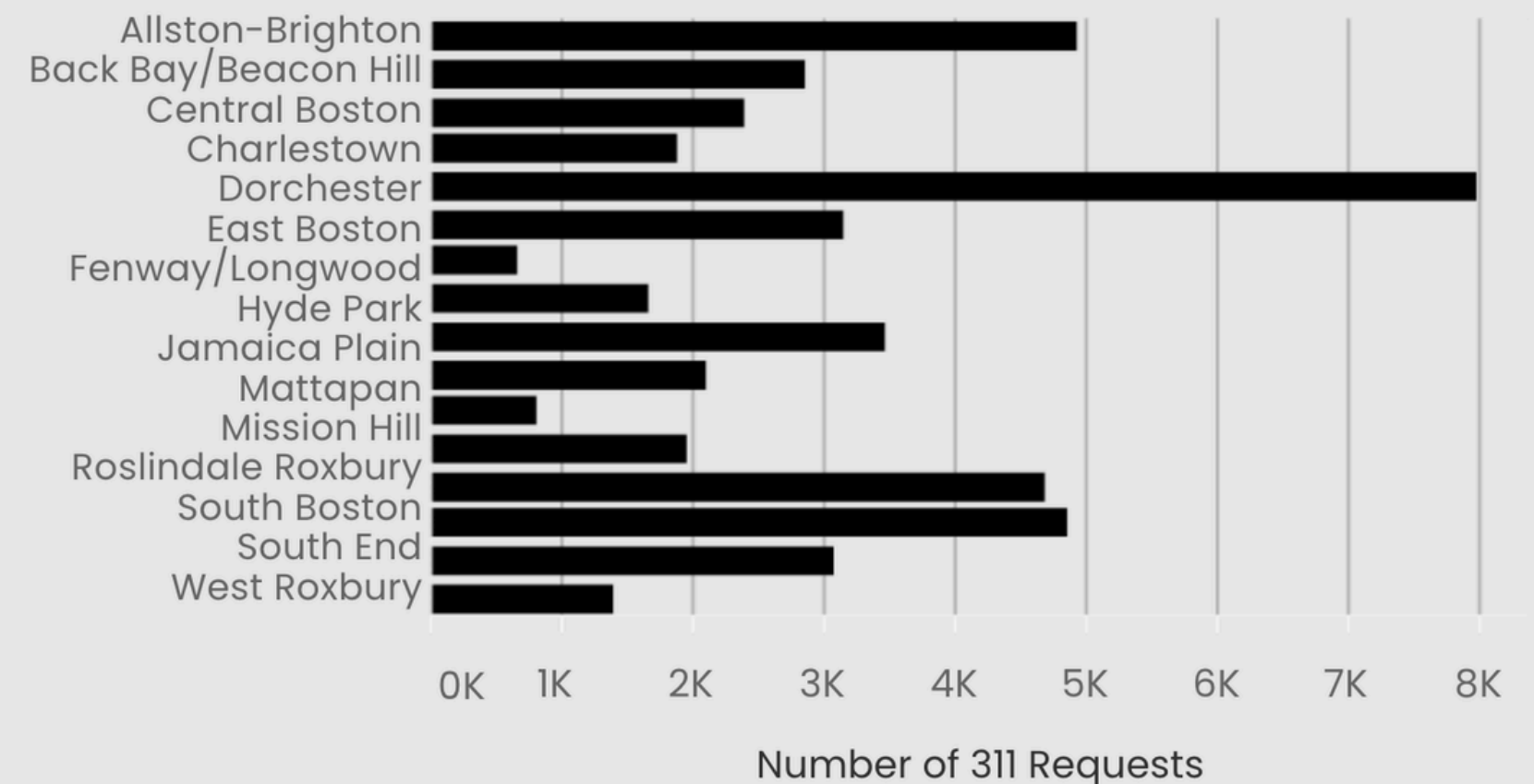
Total Open Space per Neighborhood and 311 Service Requests per Neighborhood



Total Open Space per Neighborhood

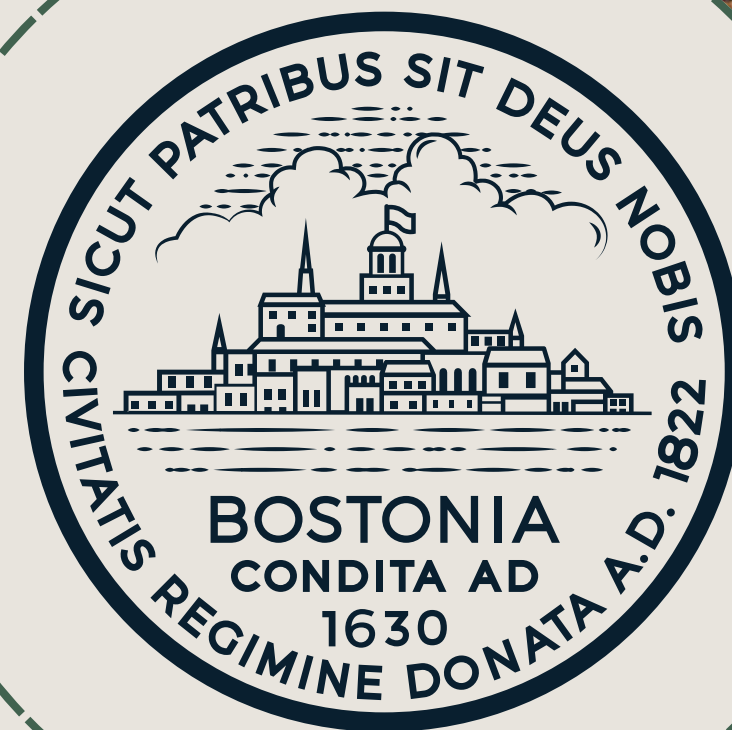


311 Service Requests per Neighborhood



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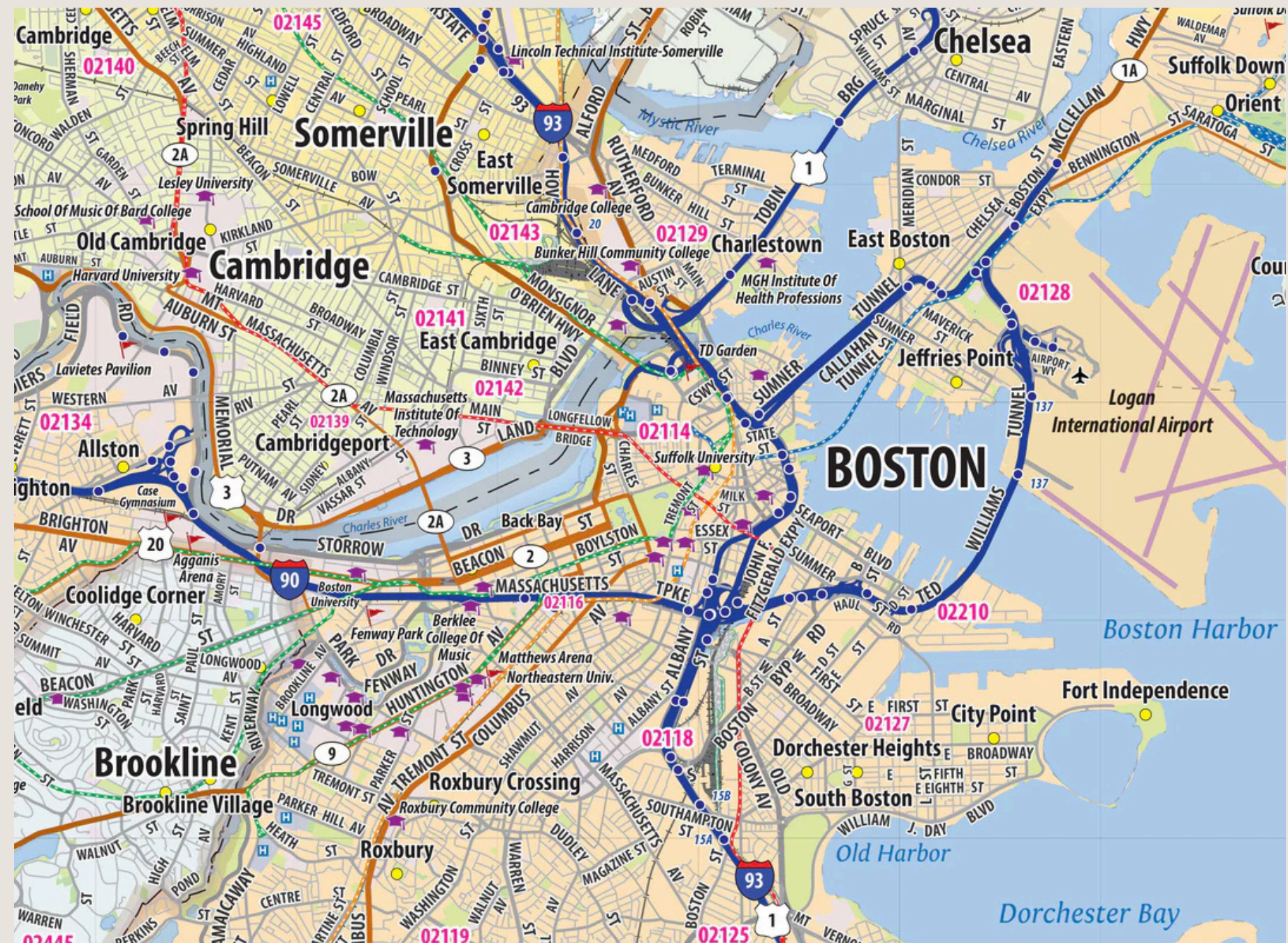
My Narrative Visualization



Conditions

Residents & Activists of Boston

- Residents who are curious about the level of responsiveness of 311 requests
- Call to Action if dissatisfactory
- People moving to Boston want to compare open space availability and 311 requests responsiveness



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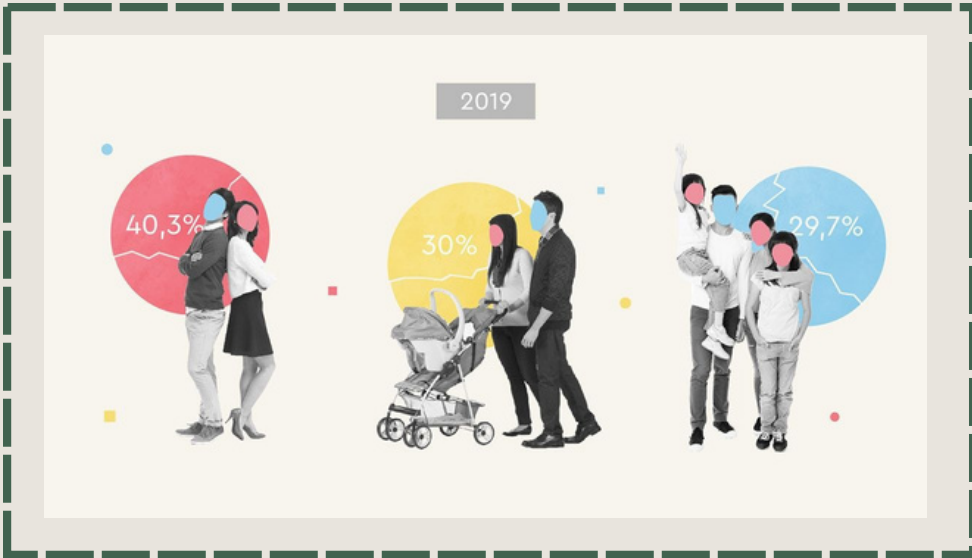
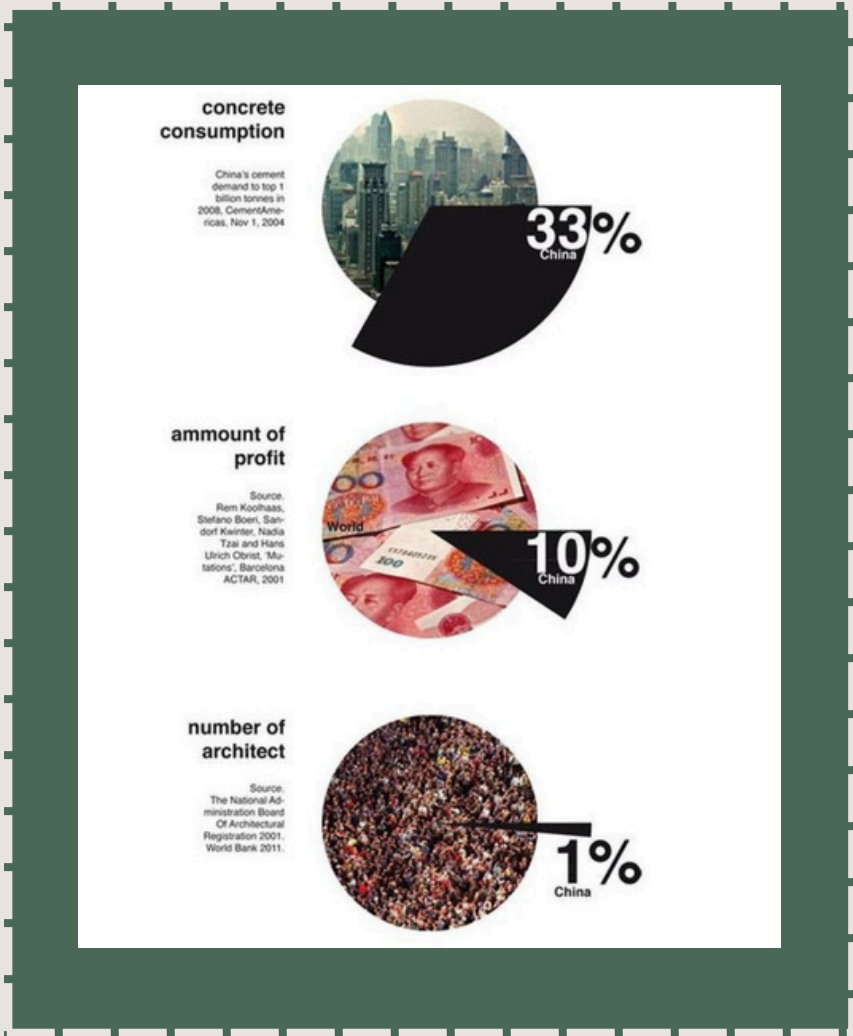
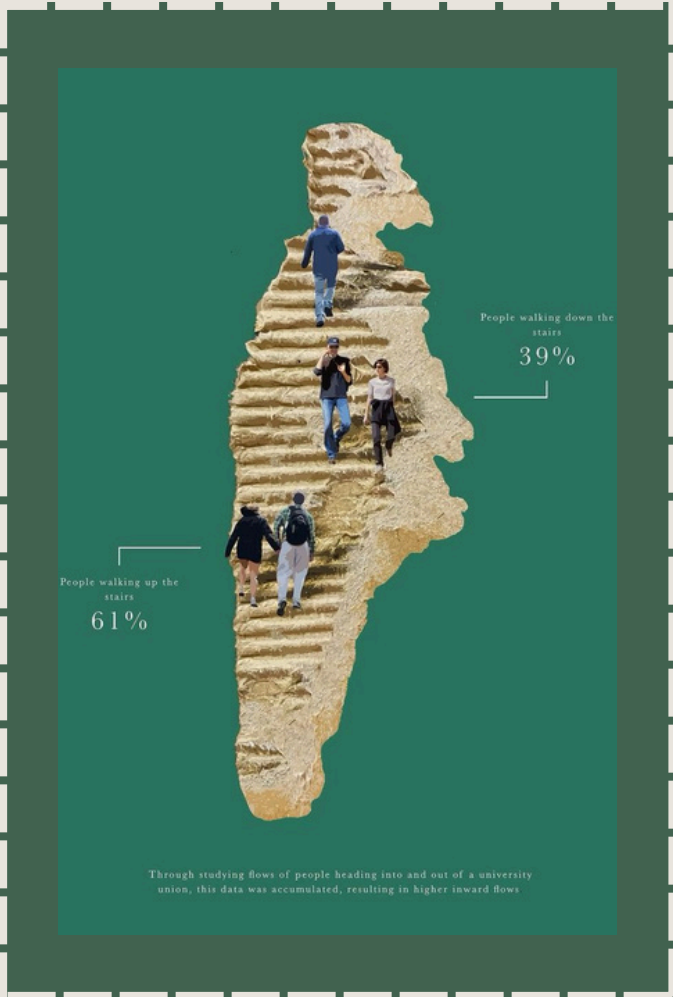
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Key References



Public Narrative

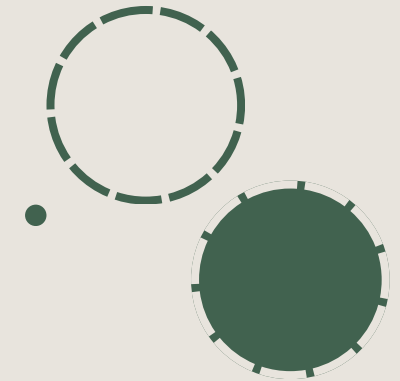


- Boston's green spaces have driven racial and economic segregation through gentrification.
- Wealthy neighborhoods continuously improve because residents reinvest in the parks that attracted them, while poorer neighborhoods never enter this cycle and stagnate.
- Top complaints (2024 citywide)
 - Parking enforcement: 63,236 requests
 - Street cleaning: 22,539
 - Improper trash barrel storage: 20,432
 - Pothole repair: 10,641
 - Missed trash/recycling: 10,594
 - Poor property conditions: 8,778
 - Tree maintenance: 8,638
- Under-resourced neighborhoods bear a heavier burden of requests.



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Project Description



What?

- A series of posters
- QR codes
- Resources links
- Contact information

Why?

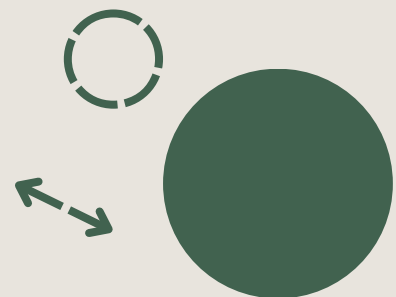
- Get Boston residents informed and engaged in local action
- Make data regarding their needs more transparent

How?

- Disperse posters with QR codes in public parks, bus stops, etc.

Themes

- Ratio of **Requests** to **Open Space Acres**
- Ratio of **Requests** to **Overdue Requests**
- Ratio of **Requests** to **Closure Rates**



Contextualized

Mock-ups

ROSLINDALE

156 Wins
More possible



3k
Total 311 Requests

156
On-time Requests

What can you do?
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

 website.com

 name@gmail.com



FENWAY

Your City Is Listening



241
Total 311 Requests

189
Requests Resolved

78.2%
Closure Rate

 website.com

 name@gmail.com



DORCHESTER

Asks the Most, Gets the Least



[https://webboston.org/](#)
[https://www.dorchester1700soco.com/](#)

 website.com

 name@gmail.com

12.6K
Total 311 Requests

400
Acres of Open Space

What can you do?
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore . adipiscing elit, sed do eiusmod tempor



JAMAICA PLAIN

122 ignored.
Which one was yours?



5K
Total 311 Requests

122
Overdue Requests

What can you do?
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

 website.com

 name@gmail.com





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Contextualized ● Mock-ups



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Thank You!

Alissa Rogozin
Spring 2026

